Wei Guo

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Research Summary

I study diffusion-based generative models and sampling algorithms, with a current focus on continuous/discrete diffusion LLMs.

Education

Georgia Institute of Technology, Machine Learning Center *Aug 2023 – Present*
Ph.D. in Machine Learning
M.S. in Electrical and Computer Engineering *May 2026*
Advisor: Professors Yongxin Chen and Molei Tao

Peking University, School of Mathematical Sciences *Sep 2019 – Jul 2023*
B.S. in Statistics
GPA: 3.848/4.0; **Rank**: 5/44
Advisor: Professor Cheng Zhang

Research Interests

- **Machine Learning**: Diffusion-based generative AI, with a focus on continuous/discrete diffusion language models and post-training (including distillation, inference-time scaling, and reinforcement learning, etc.).
- **Statistics and Probability**: Theoretical analysis and practical design of sampling algorithms (Markov chain Monte Carlo, non-equilibrium (e.g., denoising diffusion, stochastic localization) sampling, or learning-based neural samplers); Applied stochastic analysis with applications to optimal transport, stochastic optimal control, and statistical physics.

Work Experience

NVIDIA, Fundamental Generative AI Research (GenAIR) Group *May 2026 – Dec 2026*
Manager: Arash Vahdat; **Mentor**: Morteza Mardani
Research intern; working on continuous diffusion language models [8]

Preprints and Publications

(In reversed chronological order. ★ denotes selected papers. * denotes equal contribution or alphabetical order.)

[1] Jaemoo Choi, Wei Guo, Yuchen Zhu, Arash Vahdat, Molei Tao, Julius Berner, and Yongxin Chen. "Scaling Reinforcement Learning for Diffusion Models via Velocity Matching". In: *arXiv preprint arXiv:2608.23664* (2026).

[2] Jaemoo Choi*, Yuchen Zhu*, Wei Guo, Petr Molodyk, Bo Yuan, Jinbin Bai, Yi Xin, Molei Tao, and Yongxin Chen. "Rethinking the Design Space of Reinforcement Learning for Diffusion Models: On the Importance of Likelihood Estimation Beyond Loss Design". In: *Forty-third International Conference on Machine Learning*. 2026. URL: https://openreview.net/forum?id=3mJ9eTU319.

[3] Xiaochen Du, Juno Nam, Jaemoo Choi, Wei Guo, Sathya Edamadaka, Junyi Sha, Elton Pan, Yongxin Chen, Molei Tao, and Rafael Gómez-Bombarelli. "MetaDNS: Enhancing Exploration in Discrete Neural Samplers via Metadynamics". In: *Forty-third International Conference on Machine Learning*. 2026. URL: https://openreview.net/forum?id=OY7Qe2ZSx9.

[4] Wei Guo*, Jaemoo Choi*, Yuchen Zhu, Molei Tao, and Yongxin Chen. "Proximal Diffusion Neural Sampler". In: *The Fourteenth International Conference on Learning Representations*. 2026. URL: https://openreview.net/forum?id=XTHQqS7ObC.

[5] Wei Guo, Molei Tao, and Yongxin Chen. "Complexity Analysis of Normalizing Constant Estimation: from Jarzynski Equality to Annealed Importance Sampling and beyond". In: *The Fourteenth International Conference on Learning Representations*. 2026. URL: https://openreview.net/forum?id=96fJALwotm.

★ [6] Wei Guo*, Yuchen Zhu, Xiaochen Du, Juno Nam, Yongxin Chen, Rafael Gómez-Bombarelli, Guan-Horng Liu, Molei Tao, and Jaemoo Choi*. "Discrete Adjoint Schrödinger Bridge Sampler". In: *Forty-third International Conference on Machine Learning*. 2026. URL: https://openreview.net/forum?id=G9KydTWzZL.

[7] Jeongwoo Shin*, Dongsoo Shin*, Yuchen Zhu, Wei Guo, Yongxin Chen, Joonseok Lee, Jaewoong Choi, and Jaemoo Choi. "Efficient Adjoint Matching for Fine-tuning Diffusion Models". In: *The Fortieth Annual Conference on Neural Information Processing Systems (**Spotlight**)*. 2026. URL: https://openreview.net/forum?id=vSHTrySbQc.

★ [8] Zhihan Yang*, Wei Guo*, Jean-Marie Lemercier*, Simon Welker*, Yonggan Fu, Mohammad Mahdi Kamani, Sajad Norouzi, Julius Berner, Tomas Geffner, Karsten Kreis, Yongxin Chen, Molei Tao, John Thickstun, Pavlo Molchanov, Ante Jukić, Arash Vahdat, and Morteza Mardani. "Large Language Continuous Diffusion Models". In: *arXiv preprint arXiv:2610.02665* (2026).

★ [9] Zhihan Yang, Wei Guo, Shuibai Zhang, Subham Sekhar Sahoo, Yongxin Chen, Arash Vahdat, Morteza Mardani, and John Thickstun. "Continuous Diffusion Scales Competitively with Discrete Diffusion for Language". In: *The Fortieth Annual Conference on Neural Information Processing Systems*. 2026. URL: https://openreview.net/forum?id=ViiyKzhLpA.

[10] Bo Yuan, Wei Guo, and Yongxin Chen. "Reflected Proximal Dynamics for Accelerated High-Accuracy Log-Concave Sampling". In: *arXiv preprint arXiv:2610.16248* (2026).

★ [11] Yuchen Zhu*, Wei Guo*, Jaemoo Choi, Petr Molodyk, Bo Yuan, Molei Tao, and Yongxin Chen. "Enhancing Reasoning for Diffusion LLMs via Distribution Matching Policy Optimization". In: *Forty-third International Conference on Machine Learning (**Spotlight, top 2.2%**)*. 2026. URL: https://openreview.net/forum?id=09CSjVeDug.

[12] Wei Guo, Molei Tao, and Yongxin Chen. "Provable Benefit of Annealed Langevin Monte Carlo for Non-log-concave Sampling". In: *The Thirteenth International Conference on Learning Representations*. 2025. URL: https://openreview.net/forum?id=P6IVIoGRRg.

[13] Yinuo Ren*, Haoxuan Chen*, Yuchen Zhu*, Wei Guo*, Yongxin Chen, Grant M Rotskoff, Molei Tao, and Lexing Ying. "Fast Solvers for Discrete Diffusion Models: Theory and Applications of High-Order Algorithms". In: *The Thirty-ninth Annual Conference on Neural Information Processing Systems*. 2025. URL: https://openreview.net/forum?id=OuklL6Q3sO.

★ [14] Yuchen Zhu*, Wei Guo*, Jaemoo Choi, Guan-Horng Liu, Yongxin Chen, and Molei Tao. "MDNS: Masked Diffusion Neural Sampler via Stochastic Optimal Control". In: *The Thirty-ninth Annual Conference on Neural Information Processing Systems*. 2025. URL: https://openreview.net/forum?id=xIH95kXNR2.

[15] Wei Guo*, Yuchen Zhu*, Molei Tao, and Yongxin Chen. "Plug-and-play controllable generation for discrete masked models". In: *arXiv preprint arXiv:2410.02143* (2024).

Teaching Experience

- Served as a teaching assistant for the following courses at Georgia Tech:

AE 3531: Control System Analysis and Design. Lecturer: Lu Gan. *Fall 2024*
AE 8803: Optimal Transport Theory and Applications. Lecturer: Yongxin Chen. *Spring 2025*
AE 3530: System Dynamics and Vibration. Lecturer: Yongxin Chen. *Fall 2025*
AE 8803: Applied Stochastic Control and Generative Modelling. Lecturer: Yongxin Chen. *Spring 2026*

Community Service

(Numbers in parentheses indicate the number of papers reviewed. Workshop reviews not included.)

- Served as a reviewer for the following conferences:

  ICLR 2024 (1), ICLR 2025 (3), NeurIPS 2025 (5), ICLR 2026 (6), ICML 2026 (6), NeurIPS 2026 (5).

- Served as a reviewer for the following journals:

  IEEE Transactions on Information Theory (1), Transactions on Machine Learning Research (2).

Awards

PKU merit student in the academic year of 2019-2020 *Oct 2020*
PKU merit student pacesetter in the academic year of 2020-2021 *Oct 2021*
PKU merit student in the academic year of 2021-2022 *Oct 2022*
2023 Beijing Outstanding Undergraduate Graduate Award *Jul 2023*
Best poster award at Yale FDS conference "Recent Advances and Future Directions for Sampling" *Oct 2024*
2026 Georgia Tech ARC-ACO student fellowship *Jan 2026*
Gold reviewer award at ICML 2026 *May 2026*

Technical Skills

- **Mathematics and Statistics**: Markov chain Monte Carlo, convex optimization, stochastic analysis, optimal transport, applied partial differential equations.
- **Machine learning**: Diffusion/flow-based models, large language models, reinforcement learning.
- **ML systems**: Familiar with PyTorch distributed training (DDP/FSDP).
- **Programming**: Proficient in Python, including PyTorch.
- **Languages**: Chinese (Mandarin and Wu dialect, native), English (proficient), French (beginner).